

Per-agent
observations

Full soft-
attention
TopK-TarMAC

k -gate (Gumbel)
adaptive k_i

$O_i = 4$

$O_i = 3$

$O_i = 2$

$O_i = 1$

Shared
Linear

W_{qkv}
Linear

$\text{softmax}(QK^T / \sqrt{d})$
($N \times N$)

top- k mask + renorm
($N \times k$)

$m_i = \sum_{j \in T_i} \tilde{\alpha}_{ij} v_j$

Actor
 $\pi(a_i | o_i, m_i)$

